BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

First Semester 2026-2027
Course Handout

Date: 14/08/2026

In addition to Part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course No.	:	**CE F213**
Course Title	:	**SURVEYING**
Instructor In-Charge	:	**Prof. Selva Balaji M (L & T1)**
Instructor	:	**Thakur Ramjiram Singh (RS) (T2 & P1),** **Pratik Shashwat (RS) (P1 & P2), Meeniga Venkateswaralu (RS) (P1 & P2),** **Harish Kuppili (RS) (P2)**

Course Description:

The compulsory disciplinary course has been designed to introduce the basic concepts of geodesy (surveying) for Civil Engineering students. Different basic and advanced methods of measurements and traversing have been included so that the student will be able to handle a given project independently, irrespective of whether it is a road or any other infrastructure project. Important issues like curve setting, calculation of areas and volumes, which form part and parcel of any field civil engineer in his/her day-to-day activity. Hence, in this course, these areas have also been included.

Scope & Objective:

The course introduces the students to various basic techniques in surveying, viz., chain, compass, theodolite, plane table, tacheometry, traversing, and calculation of areas and volumes, along with fundamentals of a few advanced surveying techniques.

Course Learning Outcomes:

After successful completion of this course, students will be able to:

- **Understand and apply the fundamental concepts of surveying**, including methods of linear measurements and handling common field equipment such as chains, tapes, and compasses.
- **Execute and analyze compass traversing**, including the computation and correction of bearings, and apply techniques for accurate plotting and area determination.
- **Perform differential and profile levelling** and interpret level books for reduced level determination, accounting for curvature and refraction.
- **Conduct contouring operations and generate contour maps** using field measurements and appropriate plotting techniques.

- **Carry out plane table and tacheometric surveys** and apply relevant graphical and mathematical methods for plotting and distance/elevation computations.
- **Design and set out simple and transition curves** for civil infrastructure projects using chain, tape, and theodolite methods.
- **Compute areas and volumes** from cross-sections and contour maps and interpret mass haul diagrams relevant to earthworks.
- **Demonstrate knowledge of modern surveying technologies**, including total stations, photogrammetry, and remote sensing, and their practical applications.
- **Analyze sources of error in surveying measurements** and apply appropriate corrections to enhance the accuracy and reliability of data.
- **Apply theoretical knowledge in field conditions** by conducting various practical exercises such as chain surveying, curve setting, and use of total stations and GIS software.

Textbooks:

T1. Duggal S.K.; Surveying; Tata McGraw-Hill, New Delhi, Vol I and II, 4th Edition (2013).

T2. Moondra, H.S. and Gupta Rajiv, Lab Manual for Civil Engineering, CBS, 2nd edition, (2000).

Reference Books:

R1. Punmia B.C; Surveying; Laxmi Publishers, Vol I, II and III, (1990).

R2. Agor, R; A Text Book of Surveying & Levelling, Khanna Publishers. New Delhi.

R3. N N Basak: Surveying & Levelling, McGraw Hill Education (India) Private Limited, New Delhi, 2nd edition, 2014.

R4. Venkatramaiah, C; Textbook of Surveying, University Press, Second Edition, 2011.

R5. Bhavikatti, S.S; Surveying Theory & Practice, Ik International Publishing House Pvt Ltd, New Delhi, 2013.

Course Plan:

Module	Lecture No.	Topic	Topics to be covered	Text Book
Module 1	1	Introduction to the Basic Concepts of Surveying	Fundamental definitions and concepts	Chapter 1 T1 Vol I
	2-3	Linear Measurements and Instruments	Methods, accessories, ranging	Chapter 2 T1 Vol I
	4-6	Chain Survey	Steps in chain survey, field work, and plotting in field book, obstacles in chaining	Chapter 2 T1 Vol I
Module 2	7-9	Compass Survey	Instrument, principles, bearings	Chapter 3 T1 Vol I
	20-22	Traversing	Methods, adjustments, and plotting	Chapter 5 T1 Vol I
Module 3	10-11	Levelling	Instruments, collimation method, Rise and fall method, curvature, and refraction. Level book	Chapter 6 T1 Vol I
	12-13	Contouring	Objectives, use, methods	Chapter 9 T1 Vol I
Module 4	14-16	Plane Table Survey	Accessories, methods, errors	Chapter 8 T1 Vol I
	17-19	Tacheometric Surveying	Theory, instrument constants, methods	Chapter 7 T1 Vol I
Module 5	23-27	Curve Ranging	Types, properties, circular and transition curves	Chapter 11 T1 Vol I
Module 6	28-30	Trigonometrical Levelling	Geodetic survey, visibility between two places	Chapter 1 T1 Vol II
Module 7	31-33	Computation of Areas	Different methods, approximate method, planimeter	Chapter 12 T1 Vol I
	34-37	Computation of Volumes	Level section, multilevel section, volume from contour plan, mass-haul diagram	Chapter 13 T1 Vol I
Module 8	38-40	Advanced Topics	Total Stations and other advancements in surveying, Errors & adjustments, Photogrammetry, and remote sensing	Chapter 10 T1Vol-I and Chapter 3, 5, 6,7,8,9 T1 Vol II

Module	Lecture No.	Topic	Topics to be covered	Text Book
	41-42	Practical aspects of field work	Common mistakes in the field, Setting out Works	Chapter 14 T1 Vol I

Module Learning Objectives:

- To familiarize the basic fundamentals of surveying and linear measurements **[Module 1]**
- To compute and correct the measured bearings to determine area of field for different types of traverse **[Module 2]**
- To determine the reduced levels and also to establish various contours **[Module 3]**
- To plot the area of field using plane table surveying using the surveying principles and also the use of principle of tacheometry **[Module 4]**
- To set the simple circular curves for various geometric parameters **[Module 5]**
- To evaluate the errors and corrective measures due to the curvature, refraction and collimation **[Module 6]**
- To determine the areas and volumes for different road cross sections and from contour maps **[Module 7]**
- To acknowledge the recent advancements and field innovations in land surveying **[Module 8]**

Practical:

No.	Name of experiment	Classes
1	Chain Surveying and Obstructions	2
2	Compass Traversing	1
3	Profile Levelling	1
4	Contour Survey by Square Grids	1
5	Practice Session on Plane Table Surveying	1
6	Simple Circular Curve by Chain and Tape	1
7	Simple Circular Curve by Theodolite	1
8	Transition Curve by Theodolite	1
9	Height of Tall Objects by Two-Plane Method	1
10	Total Station Capabilities	1
11	Exploring Micro-Survey Software	1
12	Demonstration on GIS Software	1

Evaluation Scheme:

Sl No.	Evaluation Component	Duration	Weightage (%)	Date & Time	Nature of Component
1	Mid-semester Exam	90 minutes	25	As per the timetable	Closed book
2	Comprehensive Exam	3 hours	35	As per the timetable	Closed Book
3	Tutorials (Best 3 out of 5)	50 minutes	15	Will be notified	Open/Closed Book
4	Lab Quiz	-	5	Will be notified	Open/Closed Book
5	Lab component with Lab Record (handwritten)	-	10	Each week	Open Book
6	Class Quizzes	-	10	Every class	Open/Closed Book

Total marks for evaluation: 300 marks.

Chamber Consultation Hour: Wednesday 3:30 PM – 4:30 PM (Please get an appointment through e-mail:

Selva.balaji@pilani.bits-pilani.ac.in). Chamber location for meeting: 1212-H.

Notices: Communications will be sent only through e-mail/Nalanda/Google Classroom.

Make-up Policy:

No make-up exams will be conducted for quizzes or project presentations. Make-up request for examination must be notified before the examination, and decision will be made as per institute norms. Make-up requests made on medical grounds may not be approved unless recommended by CMO BITS Pilani.

Selva Balaji M
Instructor-in-charge
CE F213